Md Shadab Alam

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Academic Profile

PhD researcher in Industrial Design at Eindhoven University of Technology with expertise in machine learning, computer vision, reinforcement learning, and behaviour modelling for autonomous and human centred mobility systems. Published in peer reviewed venues across AI, autonomous systems, and human factors, with experience building large scale datasets, reproducible research pipelines, and interdisciplinary collaborations. Seeking Assistant Professor positions to contribute through research, teaching, student mentorship, and academic service.

Research Interests

Machine learning; computer vision; data centric AI; reinforcement learning and control; autonomous systems; human AI interaction; road user behaviour modelling; synthetic data generation; domain adaptation; simulation based evaluation; traffic safety and mobility.

Teaching Interests

Machine learning; computer vision; deep learning; reinforcement learning; robotics and autonomous systems; data science; research methods; experimental design.

Academic and Research Appointments

2023 – Present **PhD Researcher**, Eindhoven University of Technology (TU/e), Eindhoven, the Netherlands.
Conducting doctoral research on road user behaviour modelling, computer vision, and data centric machine learning for autonomous mobility and human centred transport systems.
Established the CROWD dataset from 18,000 hours of video spanning 6,798 cities and 238 countries and territories, creating a benchmark for global and cross cultural traffic behaviour analysis.
Analysed pedestrian behaviour at scale across 2,495 cities in 233 countries and territories, reporting an average crossing hesitation time of 3.18 seconds and an average crossing speed of 1.20 m/s.
Built reproducible experimentation pipelines with MLflow, DVC, Docker, seeded evaluation, and ablation studies to support traffic safety benchmarking, scenario generation, and autonomous driving validation.

2025 **Machine Learning Engineer**, Cesare, Paris, France.
Integrated PPO based reinforcement learning into the MASA SWORD simulation environment to train autonomous agents in scenario driven settings.
Structured two sided self play, seed controlled evaluation, and automated training and reporting workflows for reproducible experimentation.

Academic and Research Appointments (continued)

2021 – 2023 **Junior Research Fellow**, Centre for Industrial Consultancy and Sponsored Research, Indian Institute of Technology Madras, Chennai, India.
Conducted research on autonomous ship navigation and control, with a focus on deep reinforcement learning for underactuated autonomous surface vessels with 3 DOF KCS dynamics.
Developed DQN, DDPG, and PPO controllers for path following, wind disturbance handling, and obstacle avoidance.
Benchmarked reinforcement learning policies against PD + ILOS guidance, achieving 0.2969L RMSE on ellipse tracking and 0.4557L RMSE on figure eight tracking, with lower simulation error than the PD baseline.
Designed a hybrid PD + DRL controller and validated performance in simulation and on a 1:75.5 scale KCS platform with IMU and GPS sensing.

Research Visits

May 2024 **Leipzig University**, Leipzig, Germany.
Research visit with **Dr. Patrick Ebel** focused on computational interaction, human behaviour modelling, and mobility oriented machine learning in the context of human centred transport systems.

Mar – Apr 2024 **University of Stuttgart**, Stuttgart, Germany.
Collaborated with **Prof. Dr. Wolfram Remlinger** on human centred mobility, vehicle interior design, and user focused interaction in future transport systems.

Education

2026 **Doctoral research exchange during PhD**, École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland.
Project: *Unified Visual Perception in the Wild: A Multi Task Study on the Global CROWD Dataset.*

2023 – 2026 (expected) **PhD, Industrial Design**, Eindhoven University of Technology, Eindhoven, the Netherlands.
Thesis: *Connected data from autonomous vehicles and road users to enhance safety.*

Jun 2024 **10th International Summer School on AI and Big Data**, ScaDS.AI Dresden/Leipzig, Leipzig, Germany.
Participated in an international summer school covering generative AI, responsible AI, mathematical foundations of AI, and applied AI and big data topics.

2021 – 2023 **MSc by Research, Robotics and Artificial Intelligence**, Indian Institute of Technology Madras, Chennai, India.
Thesis: *Data driven control for marine vehicle maneuvering.*

2016 – 2020 **Bachelor of Technology, Mechanical Engineering**, Jamia Millia Islamia, Delhi, India.
Thesis: *Design of product prototype for inventory management 4.0.*

Research Publications

Journal Articles

1. **M. S. Alam** and P. Bazilinskyy, "Nineteen Years of ASMR on YouTube: A Multilingual, Theme-Level Analysis of 42,268 Videos," *Under Review*, 2026.
2. **M. S. Alam**, P. Singh, and P. Bazilinskyy, "A Survey of Day-Night Illumination Domain Translation for Outdoor Vision Covering 30 Methods 22 Datasets and Evaluation Protocols," *Under Review*, 2026.
3. **M. S. Alam**, V. Deogaonkar, and A. Somayajula, "Enhancing Autonomy: Comparative Analysis of Machine Learning-Based Controllers in Extended Action States," *Under Review*, 2025.
4. **M. S. Alam**, D. Dey, M. H. Martens, and P. Bazilinskyy, "You'll never walk alone: Inter-pedestrian distance, eHMIs, and crossing decisions in virtual reality," *Under Review*, 2025.
5. **M. S. Alam** and I. Carlucho, "Harnessing traditional controllers for fast-track training of deep reinforcement learning control strategies," *Journal of Marine Engineering & Technology*, pp. 1–12, 2024. DOI: `https://doi.org/10.1080/20464177.2024.2367276`.
6. R. Deraj, R. S. Kumar, **M. S. Alam**, and A. Somayajula, "Deep reinforcement learning based controller for ship navigation," *Ocean Engineering*, vol. 273, p. 113 937, 2023. DOI: `https://doi.org/10.1016/j.oceaneng.2023.113937`.

Conference Proceedings

1. **M. S. Alam** and P. Bazilinskyy, "Cross or Nah? LLMs Get in the Mindset of a Pedestrian in front of Automated Car with an eHMI," in *Adjunct Proceedings of the 17th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, ser. AutomotiveUI '25 Adjunct, New York, NY, USA: Association for Computing Machinery, 2025, pp. 19–34, ISBN: 9798400720147. DOI: `10.1145/3744335.3758477`.
2. **M. S. Alam**, M. Martens, and P. Bazilinskyy, "Generating Realistic Traffic Scenarios: A Deep Learning Approach Using Generative Adversarial Networks (GANs)," 3, vol. 3, AHFE Open Access, 2025. DOI: `10.54941/ahfe1005927`.
3. **M. S. Alam**, M. H. Martens, and P. Bazilinskyy, "Pedestrian Planet: What YouTube Driving from 233 Countries and Territories Teaches Us About the World," in *Proceedings of the 17th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, ser. AutomotiveUI '25, Brisbane, QLD, Australia: Association for Computing Machinery, 2025. DOI: `10.1145/3744333.3747827`.
4. **M. S. Alam**, S. H. Parmar, M. H. Martens, and P. Bazilinskyy, "Deep Learning Approach for Realistic Traffic Video Changes Across Lighting and Weather Conditions," in *2025 8th International Conference on Information and Computer Technologies (ICICT)*, Hilo, HI, USA, 2025, pp. 180–185. DOI: `10.1109/ICICT64582.2025.00034`.
5. **M. S. Alam**, Z. Wang, L. Zhang, and P. Bazilinskyy, "Exploring Veo 3's Capabilities for Generating Urban Traffic Scenes in 76 Cities Worldwide," 2025. URL: `https://www.researchgate.net/publication/393086985`.
6. P. Bazilinskyy, **M. S. Alam**, and R. Merino-Martınez, "Pedestrian crossing behaviour in front of electric vehicles emitting synthetic sounds: A virtual reality experiment," in *Proceedings of 54th International Congress Exposition on Noise Control Engineering (INTER-NOISE)*, São Paulo, Brazil, 2025. DOI: `10.3397/IN_2025_1076086`.
7. P. Bazilinskyy, **M. S. Alam**, and R. Merino-Martınez, "Psychoacoustic assessment of synthetic sounds for electric vehicles in a virtual reality experiment," in *Proceedings of 11th Convention of the European Acoustics Association (Euronoise)*, Malaga, Spain, 2025. DOI: `https://www.doi.org/10.61782/fa.2025.0710`.

8 M. Fortes-Ferreira, **M. S. Alam**, and P. Bazilinskyy, "Vibe Coding in Practice: Building a Driving Simulator Without Expert Programming Skills," ser. AutomotiveUI '25 Adjunct, New York, NY, USA: Association for Computing Machinery, 2025, pp. 60–66, ISBN: 9798400720147. DOI: 10.1145/3744335.3758482.

9 **M. S. Alam**, T. Subramanian, M. Martens, W. Remlinger, and P. Bazilinskyy, "From A to B with Ease: User-Centric Interfaces for Shuttle Buses," in *Adjunct Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, ser. AutomotiveUI '24 Adjunct, Stanford, CA, USA: Association for Computing Machinery, 2024, pp. 111–116, ISBN: 9798400705205. DOI: 10.1145/3641308.3685033.

10 **M. S. Alam**, S. K. Ramkumar Sudha, and A. Somayajula, "AI on the Water: Applying DRL to Autonomous Vessel Navigation," in *Proceedings of the Sixth International Conference in Ocean Engineering (ICOE2023)*, 2023. DOI: https://doi.org/10.48550/arXiv.2310.14938. arXiv: 2310.14938 [eess.SY].

11 J. Jose, **M. S. Alam**, and A. S. Somayajula, "Navigating the Ocean with DRL: Path following for marine vessels," in *Proceedings of the Sixth International Conference in Ocean Engineering (ICOE2023)*, 2023. DOI: https://doi.org/10.48550/arXiv.2310.14932. arXiv: 2310.14932 [eess.SY].

12 S. K. Ramkumar Sudha, **M. S. Alam**, B. Reddy, and A. S. Somayajula, "Comparison of path following in ships using modern and traditional controllers," in *Proceedings of the Sixth International Conference in Ocean Engineering (ICOE2023)*, 2023. arXiv: 2310.14940 [eess.SY]. URL: https://arxiv.org/abs/2310.14940.

Academic Service and Leadership

Peer Review
2024 – Present: Conference and journal reviewer for NeurIPS; International Journal of Control, Automation, and Systems; Journal of Intelligent & Robotic Systems; Ocean Engineering; Engineering Applications of Artificial Intelligence; and Transportation Research Interdisciplinary Perspectives.
Professional Service
2024 – 2025: Web Chair, IEEE RO-MAN 2025 conference.
2022 – 2023: Placement Coordinator, IIT Madras, supporting master's and PhD student placement activities.

Awards, Grants, and Recognition

2026: **Recipient, FAIR Data Fund (5th edition)**, 4TU.ResearchData, for FAIRification of the TraffCOCO traffic dataset.
2026: **Recipient, NWO/SURF Small Computing Time allocation**, national computing facilities support for research computing.
2025: **Honorable Mention, AutomotiveUI '25**, for the paper *Pedestrian Planet: What YouTube Driving from 233 Countries and Territories Teaches Us About the World*.
2022: **Winner, VRX Competition**, organised by the Office of Naval Research, the Naval Postgraduate School, and Open Robotics.
2022: **Winner, Data Analysis Hackathon**, IEEE Oceans 2022.

Technical Areas

Programming: Python, C++, C, SQL, R. **Machine learning and AI:** PyTorch, TensorFlow, JAX, NumPy, Pandas, transformers, reinforcement learning, GANs, graph neural networks, computer vision. **Research infrastructure:** MLflow, DVC, Docker, Kubernetes, CI/CD for ML, SLURM. **Data and visualisation:** Apache Spark, Matplotlib, Plotly, Dash, Power BI. **Robotics and simulation:** ROS, sensor integration, Unity, Unreal Engine.